Technological Origin of Labor Adjustments: Horizontal versus Vertical Innovation

Capital Theory

October 5, 2021

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Two Types of Incumbent Innovation

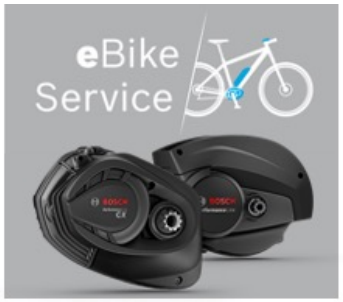
VS

Vertical Innovation

Generations of Hand Drills by Bosch

- Replaces its own product
- Small labor churning

BOSCH +



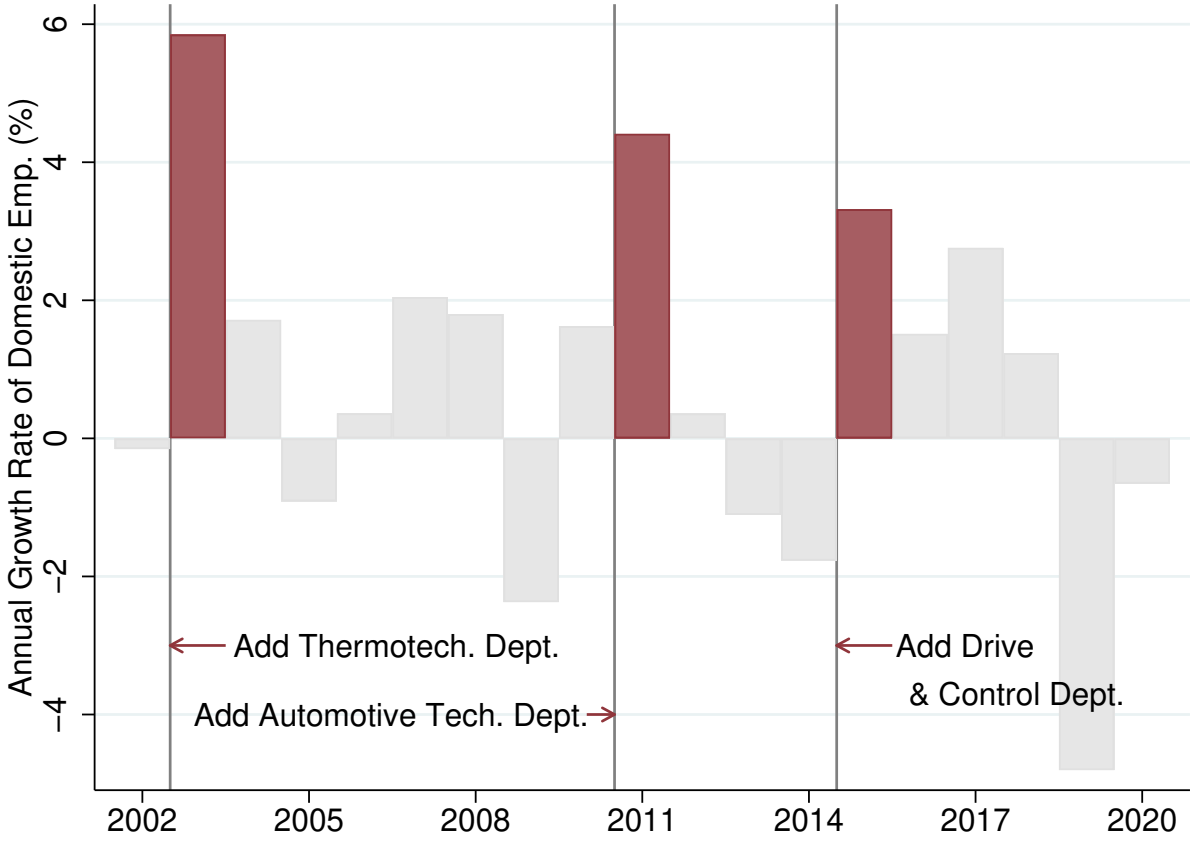
Horizontal Innovation

Bosch's expansion to E-Bike Motors

- Replaces others' products
- Large labor churning

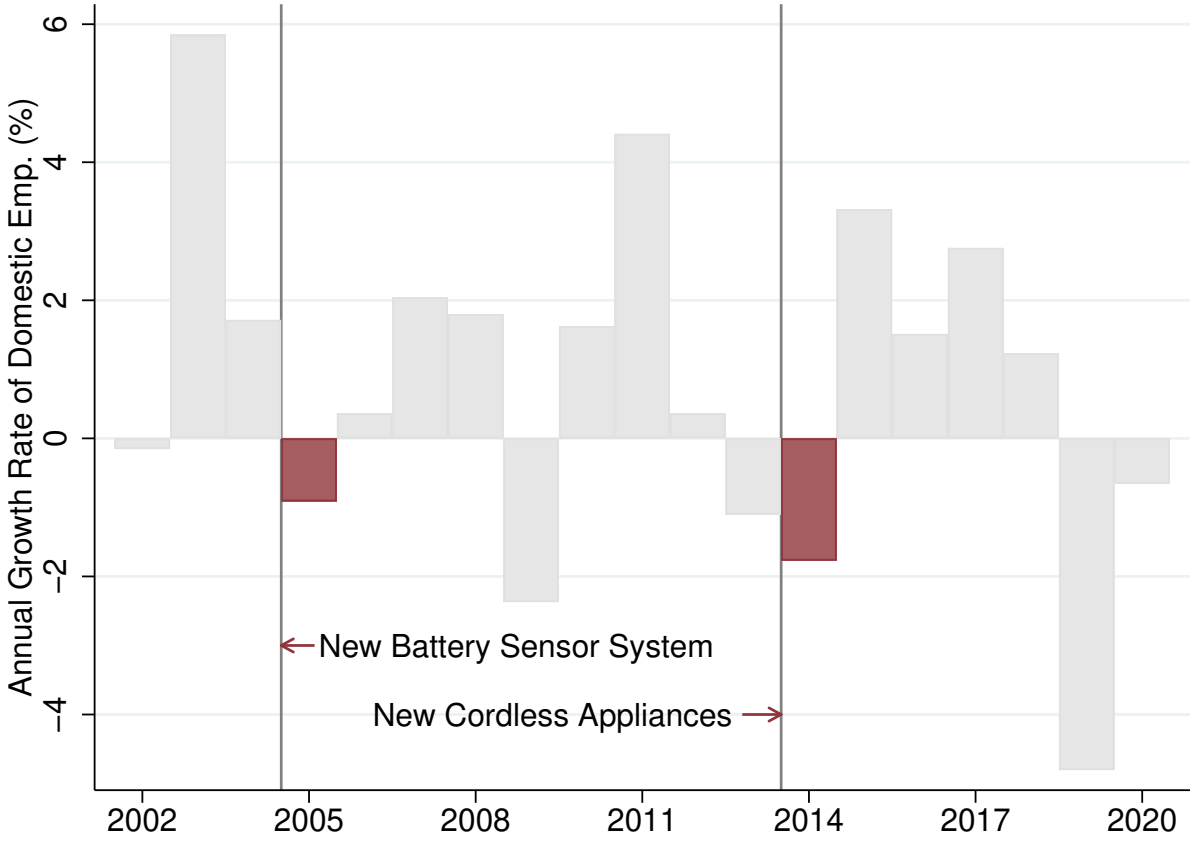
Employment of Bosch

- Firm-level employment grows with horizontal expansion.



Employment of Bosch

- Firm-level employment does not grow with vertical improvements.



This paper: Labor Market Churning and Growth

- Labor reallocation from low to high productivity firms is an essential component of productivity growth.
- Consequently, labor market churning is often used as a diagnosis tool for aggregate growth.
- This relation depends on which type of innovation is dominant in the economy.
  - Both vertical and horizontal innovations can enhance productivity.
  - Vertical innovation has smaller effects on labor market churning than horizontal innovation.
- However, this distinction is not often made within a unified framework because of difficulty of isolating these two innovations.
  - A few exceptions include Garcia-Macia, Hsieh, and Klenow (2019) and Akcigit and Kerr (2018)

Research Question

- **Research Question**:
  - How is labor market churning related to aggregate growth?
- **Explanation of This Paper**:
  - Vertical innovation has smaller effects on churning than horizontal innovation.
  - Vertical innovation accounts for most productivity growth.
  - $\Rightarrow$ Labor market churning does not respond elastically to aggregate growth.
- **Application**:
  - The Hartz labor market reform improves productivity growth and welfare additionally through more vertical innovation.
  - Job turnover rates respond less elastically to the labor market reform.

Identifying Incumbent Innovation through Survey Questions

- **Horizontal Innovation**:
  - Did your establishment start to offer a product/service that had been available on the market before?
  - ⇒ Expansion to a new product line
  - ⇒ Productivity improvement through reallocation of product lines
- **Vertical Innovation**:
  - Did your establishment improve or further develop a product or service which had previously been part of your portfolio?
  - & Did you develop or implement procedures which have noticeably improved production processes or services?
  - ⇒ Productivity improvement within a product line

German Establishment Panel Survey (IAB-BP)

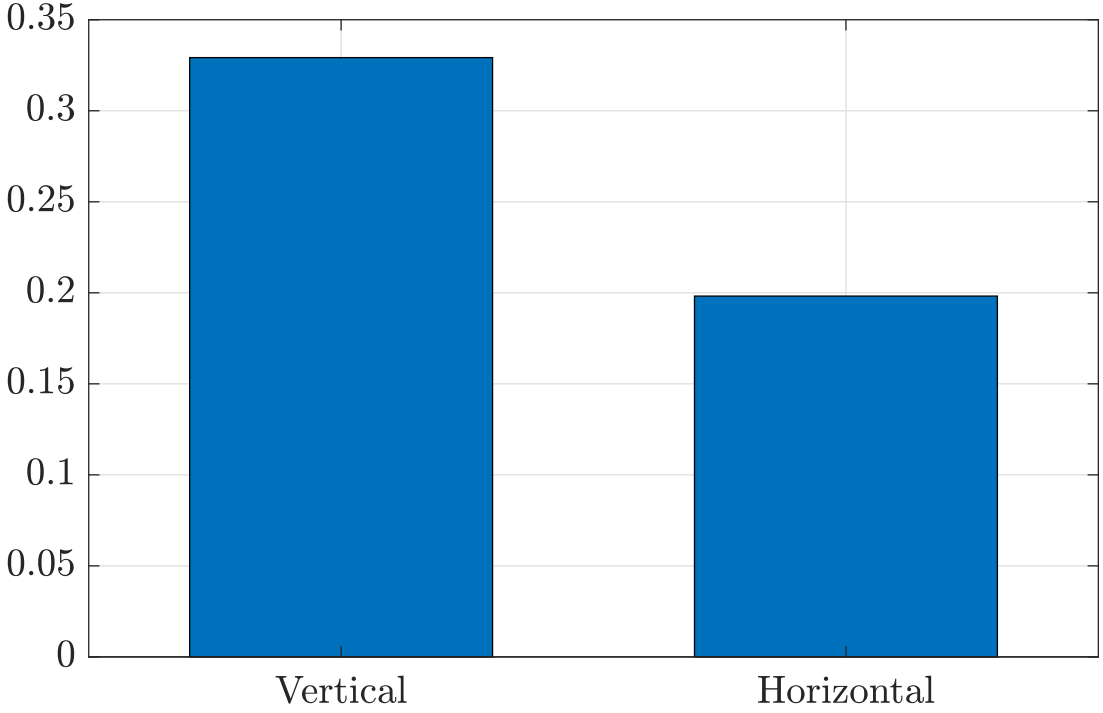
- Annual survey of establishments in Germany.
- Coverage
  - Starts in 1993 for West Germany and 1996 for East Germany, currently until 2017.
  - Representative random sample stratified according to establishment size, industry, and federal state.
  - Approximately 16,000 establishments per year.
- Innovation module started in the 2008 survey.
- Substantial changes in covered variables and questions in 2002.

  $\Rightarrow$ Restrict my samples to the post-2002 period.
- IAB-BP can be linked to the administrative employer-employee matched data.
  - Employment, Imputed daily wage of full time employees as of June 30.

Data Description

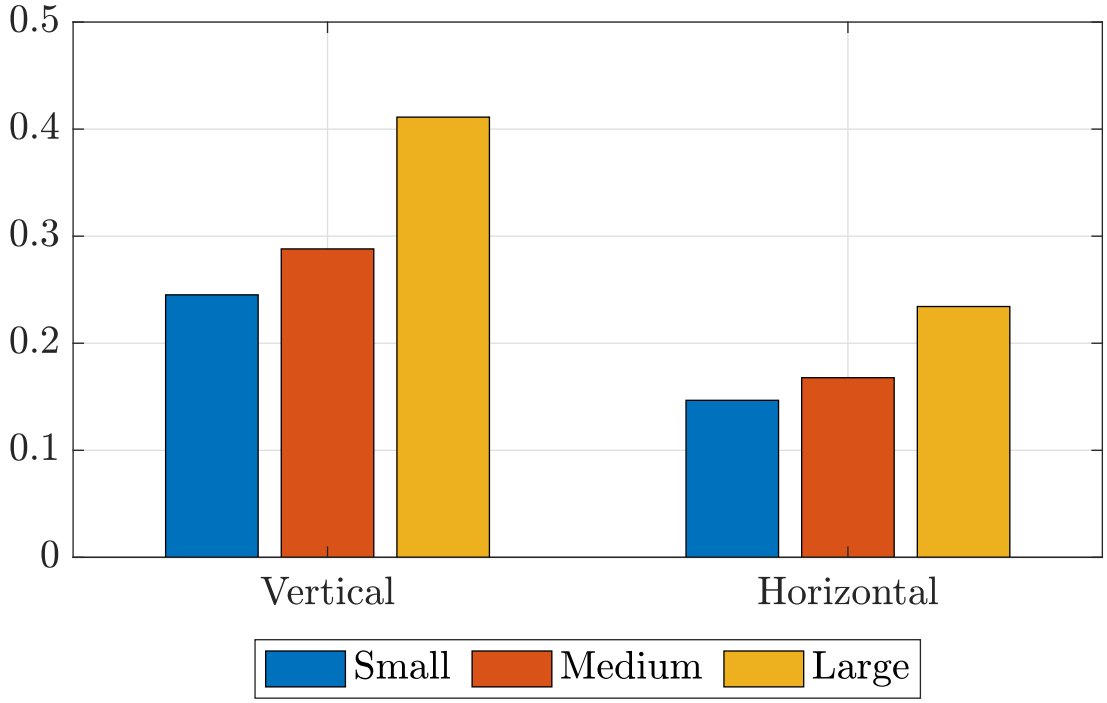
Fraction of Innovating Establishments

- More firms innovating vertically than horizontally.



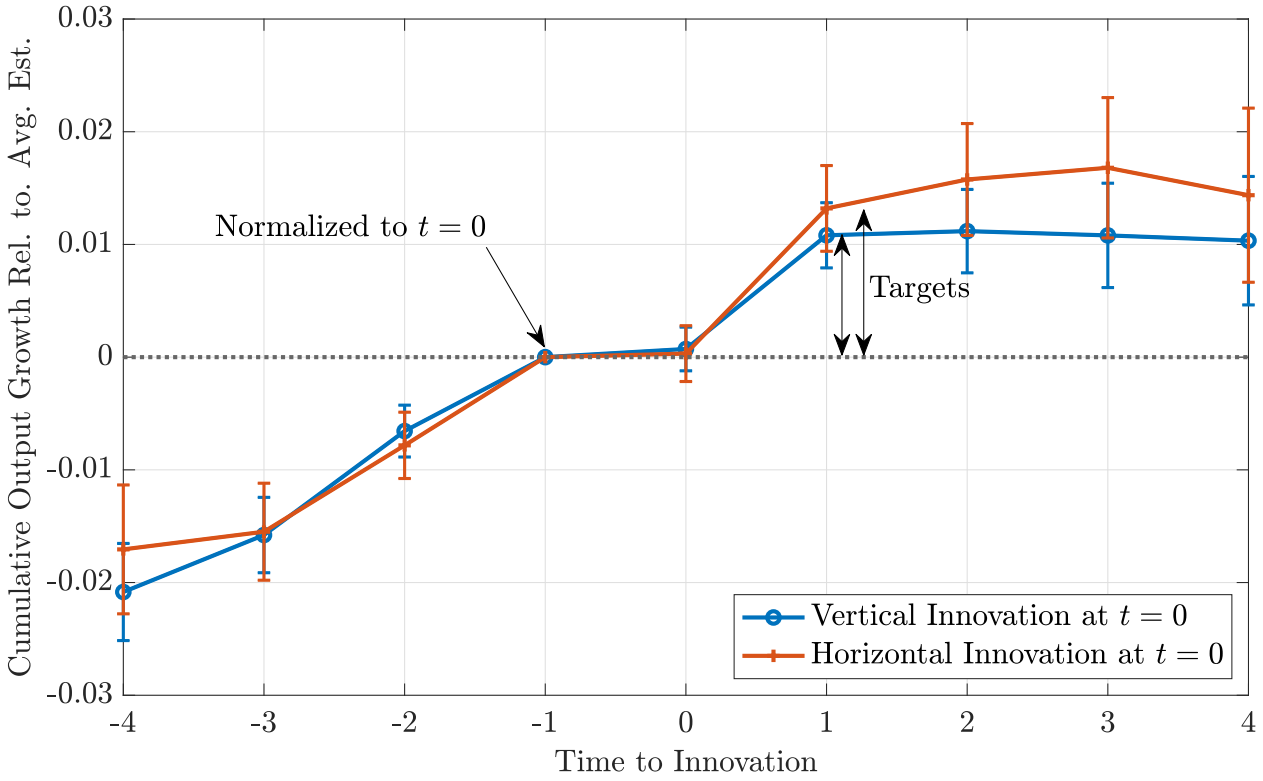
Lagged Size and Fraction of Innovating Est.

- Larger establishments are more likely to innovate.



Innovation and Establishment Growth

- Exists pre-trend but the gap widens after the innovation



Literature Review

- **Schumpeterian Growth Model**:
  - Aghion and Howitt (1992), Klette and Kortum (2004), Lentz and Mortensen (2008), Acemoglu et Akcigit (2012), Acemoglu and Cao (2015), Acemoglu et al. (2018), Akcigit and Kerr (2018), Atkeson and Burstein (2019), Akcigit, Alp, and Peters (2021), Peters (2020)
- **Technological Change and Employment**:
  - Davis and Haltiwanger (1999), Postel-Vinay (2004), Pissarides and Vallanti (2007), Hornstein, Krussel, and Violante (2007), Michau (2013), Decker et al. (2014, 2016), Coles and Mortensen (2016), Faberman (2017), Engbom (2019), Garcia-Macia et al. (2019)
- **Empirical Measures of Innovation**:
  - Kortum (1997), Aghion et al. (2005, 2009), Hashmi (2013), Seru (2014), Kogan et al. (2017), Akcigit and Kerr (2018), Dorn et al. (2020), Argente et al. (2020), Webb (2020), Zolas et al. (2021)
- **Effect of the Hartz Reform**:
  - Krause and Uhlig (2012), Launov and Wälde (2013, 2016), Bradley and Krüger (2019), Hochmuth et al. (2021), Bauer and King (2019)

Model

Model - Overview

- **Goal**
  - Construct a framework to quantify the impact of firm innovation behavior on macro outcomes such as growth and job flow rates.
  - Incorporate endogenous horizontal and vertical innovation choices into the BGP à la Acemoglu and Cao (2015) and Akcigit and Kerr (2018).
- **Agents**
  - Representative households consume final goods and supply labor.
  - Competitive final good producers combine intermediate inputs.
  - Intermediate good producers hire workers & innovate vertically and horizontally on multiple product lines.
- **Technology**
  - A product line produces intermediate inputs using labor.
  - Intermediate inputs are differentiated & aggregated with CES.

Representative Household

- The representative household maximizes the following lifetime utility.

$$\max_{C_t, L_t} \int_0^\infty e^{-\rho t} \left(\log(C_t) - \frac{\phi L_t^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} \right) dt$$

subject to

$$C_t + \dot{b}_t = W_t(1-\tau_L)L_t + \int \pi_{jt} dj + r_t b_t + \tau_L W_t \tilde{L}_t$$

- On the BGP, $C_t = Y_t$ grows at the common rate g_Y, $w_t := W_t/Y_t = w$ is constant, and the first order condition becomes

$$r_t = r = \rho + g_Y$$
$$\phi \left(L_t\right)^{\frac{1}{\eta}} = \frac{W_t(1-\tau_L)}{C_t} = w(1-\tau_L)$$

- Zero net supply of real bonds, $b_t = 0$, $\forall t$.

Final Goods Production and Factor Demand

- The final good is produced by

$$Y_t = \left(\int_0^1 a_{jt} y_{jt}^{\frac{\sigma-1}{\sigma}} dj \right)^{\frac{\sigma}{\sigma-1}},$$

where $\sigma > 1$ is the elasticity of substitution.

- An intermediate good producer solves

$$p_{jt} = \arg\max_p \left\{ y_{jt}(p)p - y_{jt}(p)W_t \right\}.$$

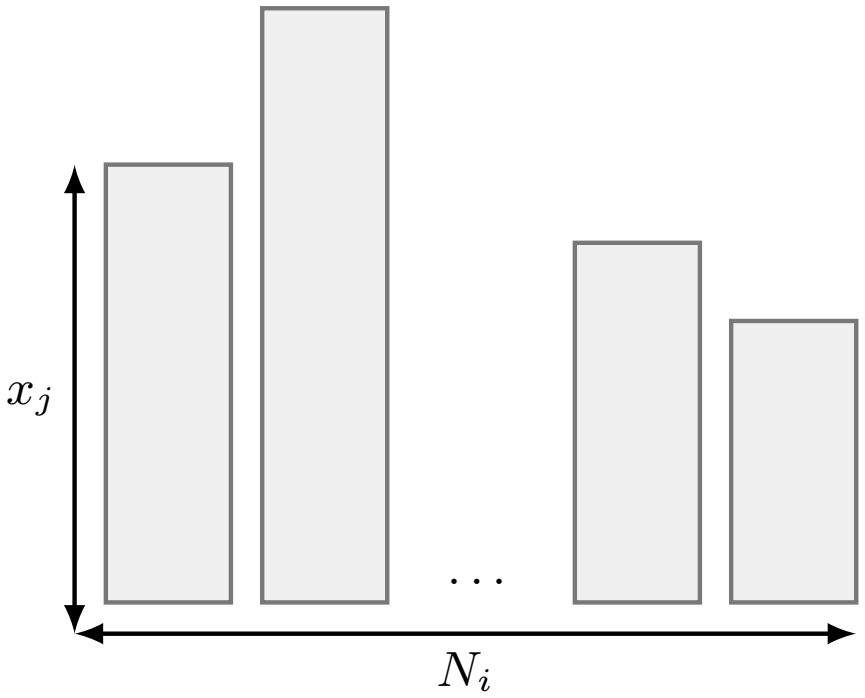
- The solution gives the following profit and production labor demand.

$$Y_t \pi(x_{jt}) := Y_t x_{jt}/\sigma, \quad W_t l^p(x_{jt}) = (\sigma - 1) Y_t x_{jt}/\sigma$$

where $x_{jt} := a_{jt}^{\sigma} / A_t^{\sigma-1}$ and $A_t := \left(\int_0^1 a_{jt}^{\sigma} dj \right)^{\frac{1}{\sigma-1}}$.

- For the zero-profit of final goods producers, $W_t = A_t \frac{\sigma-1}{\sigma}$.

Graphics: Revenue and Depreciation



- Firm state $= \{x_j\}_{j=1}^{N_i}$.
  N_i - number of product lines
  x_j - relative quality
- Without any firm-level shock,

$\Rightarrow$ x depreciates over time with A_t growing over time.

$\Rightarrow$ Nothing happens to N_i.

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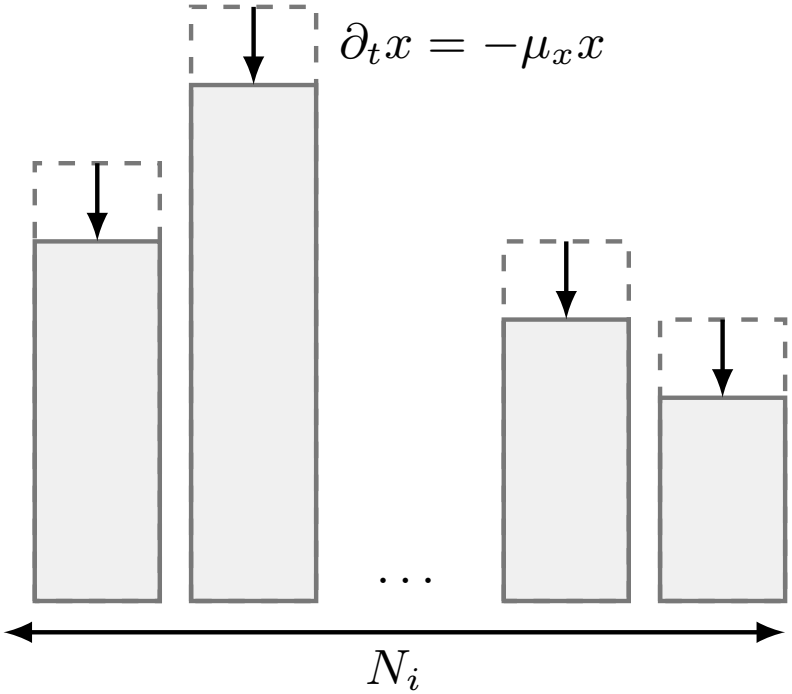

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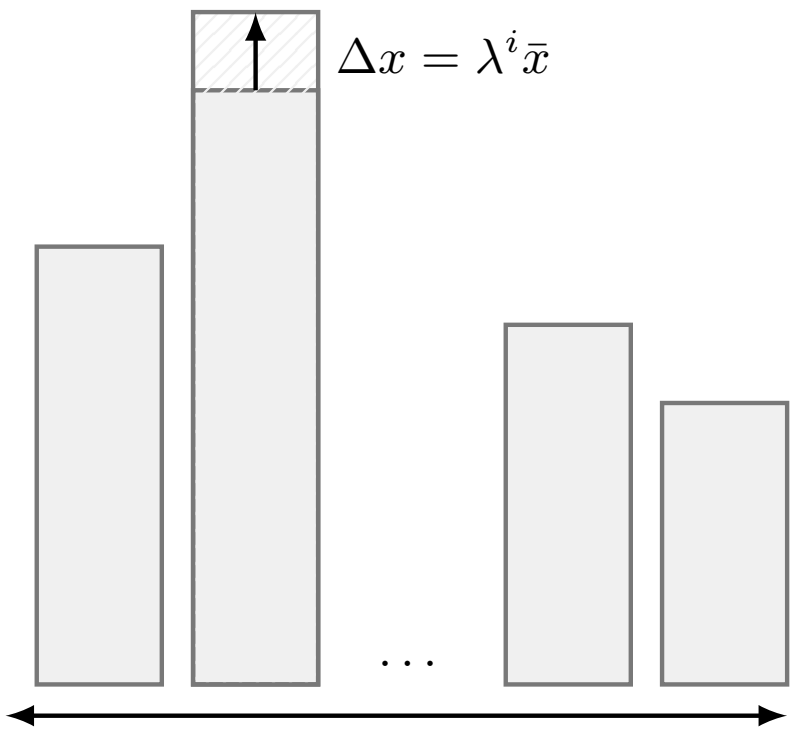
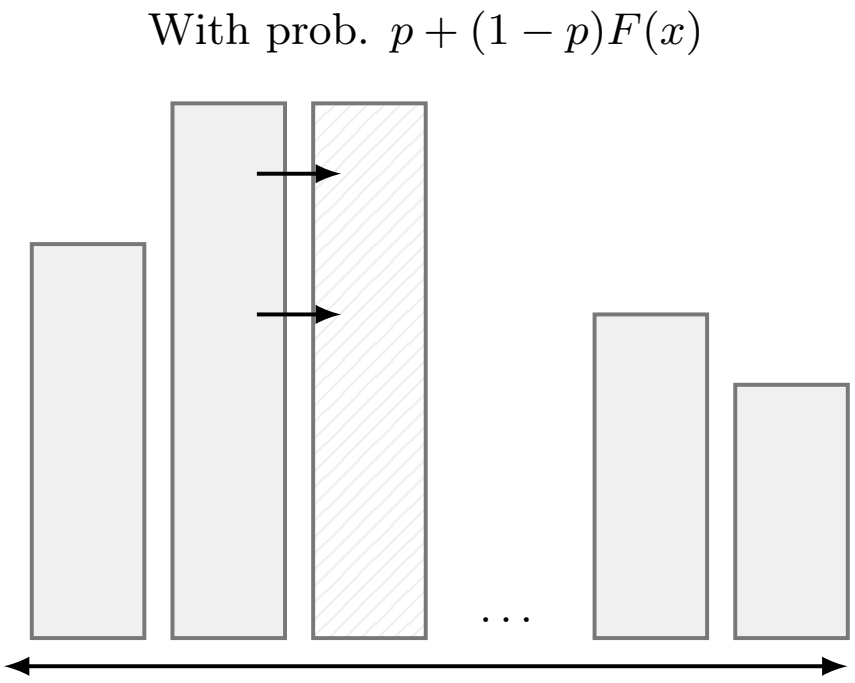
Graphics: Vertical Innovation



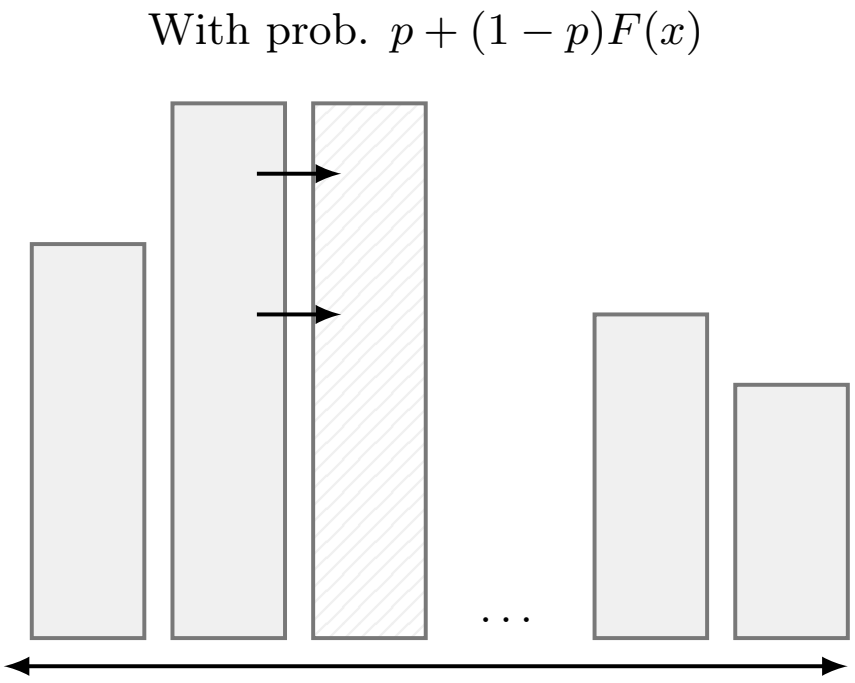
- A firm can vertically innovate with a Poisson rate ξ at the labor cost of $W_t c_\xi(\xi)$ per product line.
- $c'_\xi, c''_\xi > 0$
- Vertical innovation increases relative productivity of a product line by $\lambda^i \bar{x}$ instantaneously.
- Vertical innovation choice is product-line-specific.

Graphics: Horizontal Innovation



- A firm can horizontally innovate with a Poisson rate ζ at the labor cost of $W_t c_\zeta(\zeta)$.
- Horizontal innovation adds a new product line having the same productivity as the underlying product line.
- With probability p, it is matched to a vacant product line.

Graphics: Horizontal Innovation



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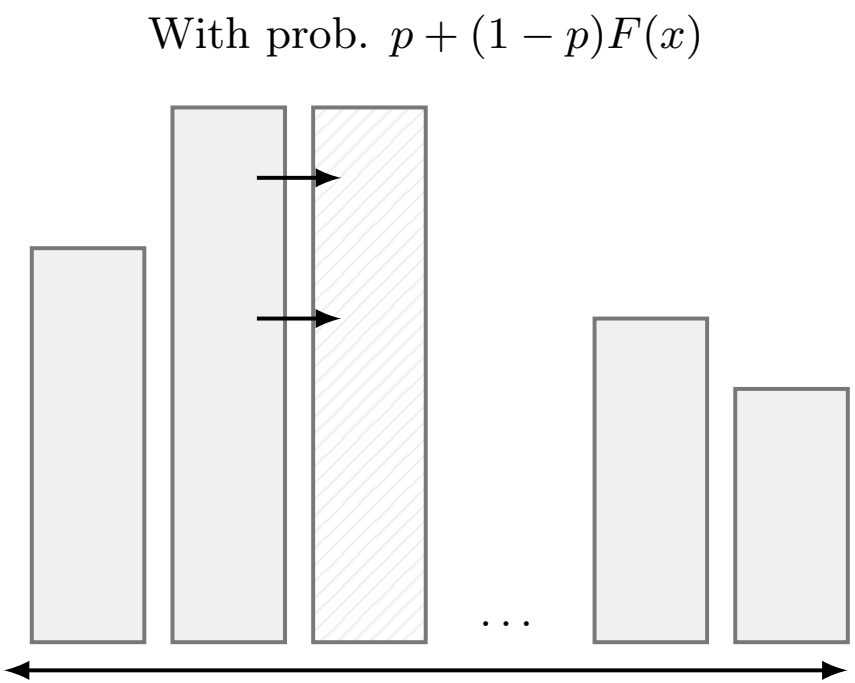
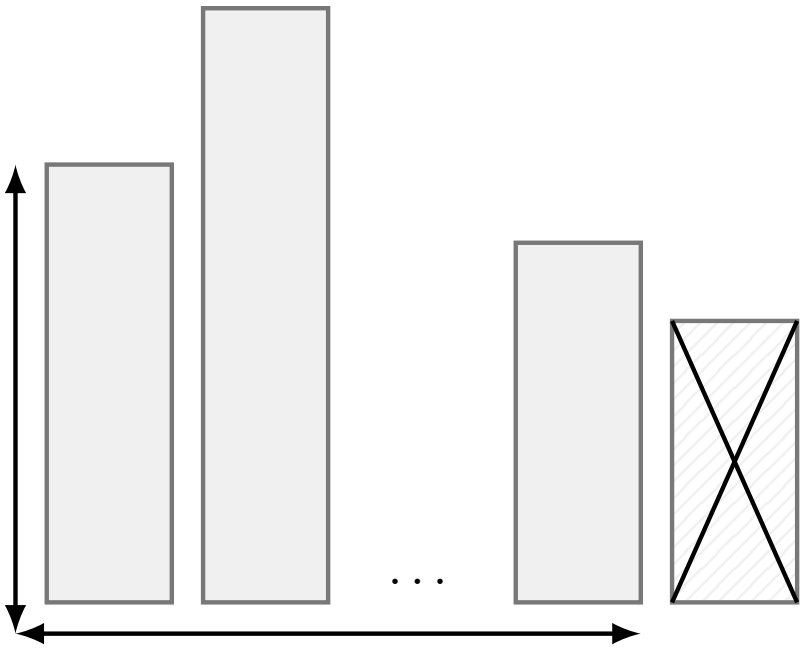


- A firm can horizontally innovate with a Poisson rate ζ at the labor cost of $W_t c_\zeta(\zeta)$.
- Horizontal innovation adds a new product line having the same productivity as the underlying product line.
- With probability $1-p$, it is matched to an occupied product line.
- It produces on matched lines as long as it is more productive than the matched firm.

Graphics: Creative Destruction



- A firm experiences a creative destruction shock on a product line with x at rate $\tau(x)$.
- Matched firms replace old firms if they are more productive.

$\Rightarrow$ $\tau'(x) < 0$.

- Horizontally expanding firms and entrants are the sources of creative destruction shock.

More Graphical Illustration

Firm Values and Stationary Value

- The profits are summed over product lines.
- Other then exogenous firm exit shocks, all decisions & shocks are independent across product lines.

$\Rightarrow$ Firm value is the sum of product line values.

$$\mathfrak{V}_t\left(\{x_j\}_{j=1}^{N_i}\right) = \sum_{j=1}^{N_i} V_t\left(x_j\right)$$

- On the BGP, all input prices, aggregate output, and profits grow at the same rate g_Y.

$\Rightarrow$ $v_t(x) := V_t(x)/Y_t$ is a time-invariant function of x.

Firm and Product Line Values

- The stationary component of a product line value is characterized by

$$
\begin{aligned}
(\rho+\delta+\tau(x))\,v(x) = \max_{\xi,\zeta\geq 0}\Big\{ & \pi(x) - g_x x v'(x) \\
& - wc_\xi(\xi) + \xi\left(v\left(x+\lambda^i\bar{x}\right) - v(x)\right) \\
& - wc_\zeta(\zeta) + \zeta\left(p+(1-p)F(x)\right)v(x) \\
& - (\tau(x)+\delta)\,\kappa x\Big\}.
\end{aligned}
$$

where wc_ξ and wc_ζ are the labor costs of vertical and horizontal innovation, respectively.

Definition of BGP Equilibrium

- The Balanced Growth Path (BGP) equilibrium is defined as value function $v(x)$; policy function $\xi(x)$, $\zeta(x)$; product line distribution $f(x)$; creative destruction rates $\tau(x)$; growth rates μ_x, μ_Y; entry rate χ; constant wage level w; a constant interest rate r; the constant labor supply L such that

  1. The representative household optimizes.
  2. Value function and policy function satisfy the HJB equations.
  3. The distribution of product lines over x stays constant. The mass of product lines is also time-invariant.
  4. The free entry condition is met to pin down entry rate, χ.
  5. Growth and creative destruction rates are time-invariant.
  6. Final goods and labor markets are cleared.

RepH Distribution Creative Destruction Free Entry & Labor Market Clearing

Innovation and Employment Change

- Firm-level employment has two components:

$$\sum_{j=1}^{N_i} l_{jt} = \sum_{j=1}^{N_i} \{ \underbrace{\frac{\sigma - 1}{\sigma} \frac{x_{jt}}{w_t}}_{\text{prod. labor}} + \underbrace{c_\zeta(\zeta(x_{jt})) + c_\xi(\xi(x_{jt}))}_{\text{R\&D labor}} \}$$

- Vertical innovation increases labor inputs through higher x_{jt}.
- Horizontal innovation increases labor inputs by an increase in N_i.
- Creative destruction decreases labor inputs by a decrease in N_i.
- Without any establishment-level shock, labor inputs decrease through depreciation on x_{jt}.

Aggregate Productivity Growth

- The growth rate on the BGP can decomposed into

$$
\begin{aligned}
g_x \bar{x} = & \underbrace{\int \left[\int_{s \in (x - \lambda^i \bar{x}, x]} M \xi(s) dF(s) \right] dx}_{\text{vertical innovation}} \\
& + p \underbrace{\int \left[\int_{s > x} M \zeta(s) dF(s) \right] dx - p \bar{\zeta} \bar{x}}_{\text{horizontal innovation (1)}} \\
& + p \underbrace{\int \left[\chi - \chi \int_{s \leq x - \lambda^e \bar{x}} dF(s) \right] dx - p \chi \bar{x}}_{\text{entrant innovation (1)}} \\
& + (1-p) \underbrace{\int \left(\int_{s > x} M \zeta(s) dF(s) \right) F(x) dx}_{\text{horizontal innovation (2)}} \\
& + (1-p) \underbrace{\int \left(\chi \int_{s > x - \lambda^e \bar{x}} dF(s) \right) F(x) dx}_{\text{entrant innovation (2)}}
\end{aligned}
$$

where $\bar{\zeta} := \int \zeta(s) dF(s)$ is the average horizontal innovation rate.

Calibration

Calibration - Step 1

- Iso-elastic innovation cost functions

$$c_\xi(\xi) = c_\xi \frac{\epsilon_\xi}{\epsilon_\xi + 1} \xi^{\frac{\epsilon_\xi + 1}{\epsilon_\xi}}$$

$$c_\zeta(\zeta) = c_\zeta \frac{\epsilon_\zeta}{\epsilon_\zeta + 1} \zeta^{\frac{\epsilon_\zeta + 1}{\epsilon_\zeta}}$$

$$c_\chi(\chi) = c_\chi \chi^{\frac{\epsilon_\chi + 1}{\epsilon_\chi}}$$

- q^v, q^h: prob. of successful innov. conditional on survey 'yes'.
- Within firm, innovation events are independent across product-lines.
- I externally calibrate two model parameters to match German macro moments.

Moment	Meaning	Value	Source
r	Discount rate	8%/12	Average Stock Market Rate of Return
δ	Exogenous exit rate	1%/12	Fackler et al. (2013)
$\bar{L}$	Labor supply in the steady state	0.38	Normalization ($w = 1$)

Calibration - Step 2

- Then, I internally match 10 moments with 10 parameters.

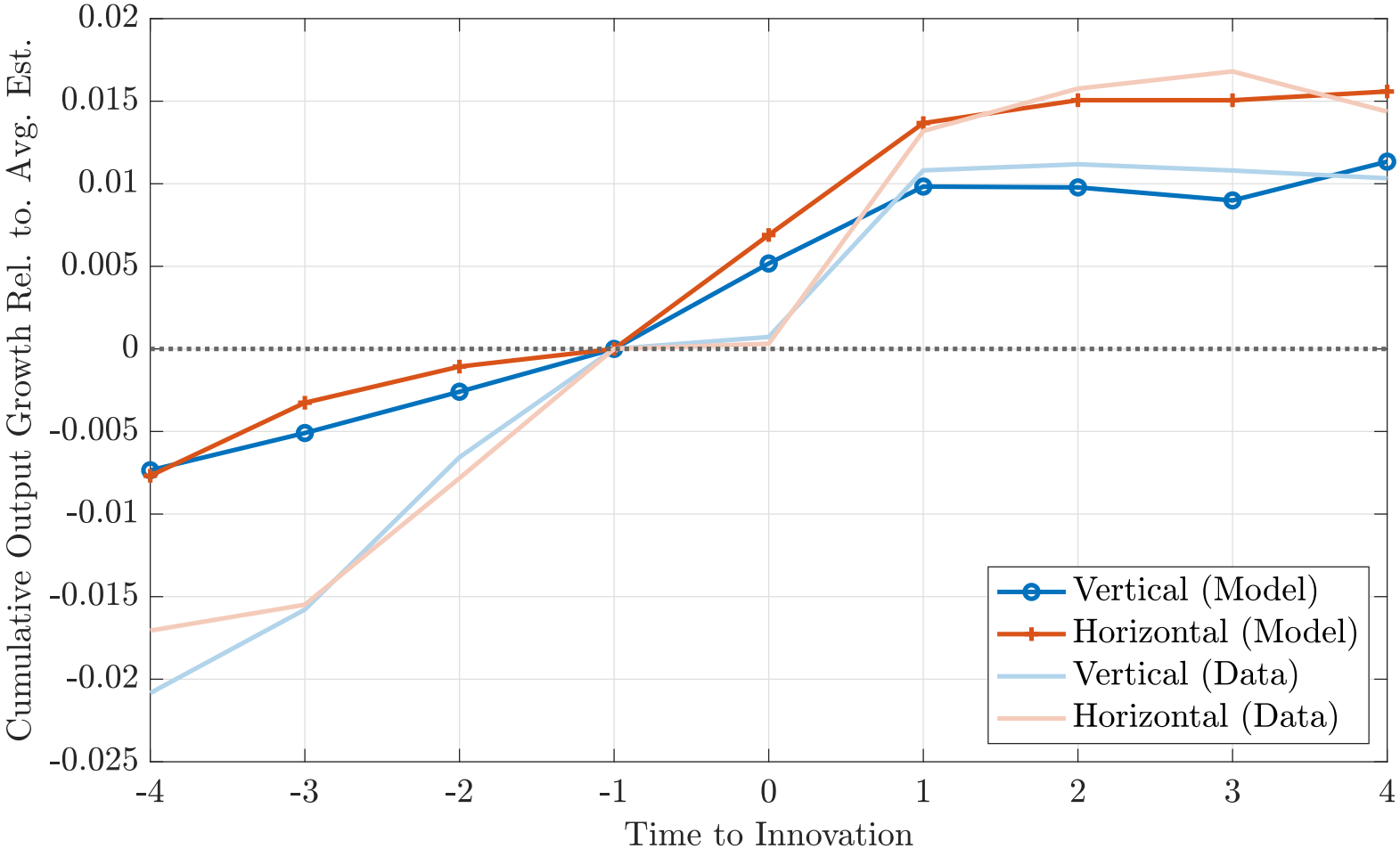
Moment	Data	Model	Target
Annual Growth Rate	1.34%	1.30%	λ^i
Frac. of Vertical Innov.	29.4%	29.1%	c_ξ
Frac. of Horizontal Innov.	17.3%	17.3%	c_ζ
Verti. Innov. Frac. Diff. across Size 2T and 1T	3.6%p	1.8%p	ϵ_ξ
Horiz. Innov. Frac. Diff. across Size 2T and 1T	2.0%p	1.4%p	ϵ_ζ
2-year Size Gr. after Verti. Innov. Rel. Avg.	1.08%p	0.98%p	q^v
2-year Size Gr. after Horiz. Innov. Rel. Avg.	1.32%p	1.37%p	q^h
2-year Size Gr. w/o Innov. Rel. Avg.	-1.55%p	-1.53%p	σ
Employment Share of Establishment Age ≤ 1	1.08%	1.51%	λ^e
Establishment Share of Establishment Age ≤ 1	1.9%	1.9%	χ

Calibration Results

	Parameter	Value
λ^i	Step Size of Vertical Innovation	2.72%
λ^e	Step Size of Entrant Innovation	3.09%
σ	Elasticity of Substitution across Prod. Lines	1.39
c_ξ	Vertical Innovation Cost Shifter	2039.84
ϵ_ξ	Vertical Innovation Cost Elasticity	0.53
c_ζ	Horizontal Innovation Cost Shifter	2.62e+06
ϵ_ζ	Horizontal Innovation Cost Elasticity	0.31
$12 \times \chi$	Annual Share of Entrant Draws	0.92%
q^h	Probability of Successful Expansion	3.12%
q^v	Probability of Successful Improvement	55.00%

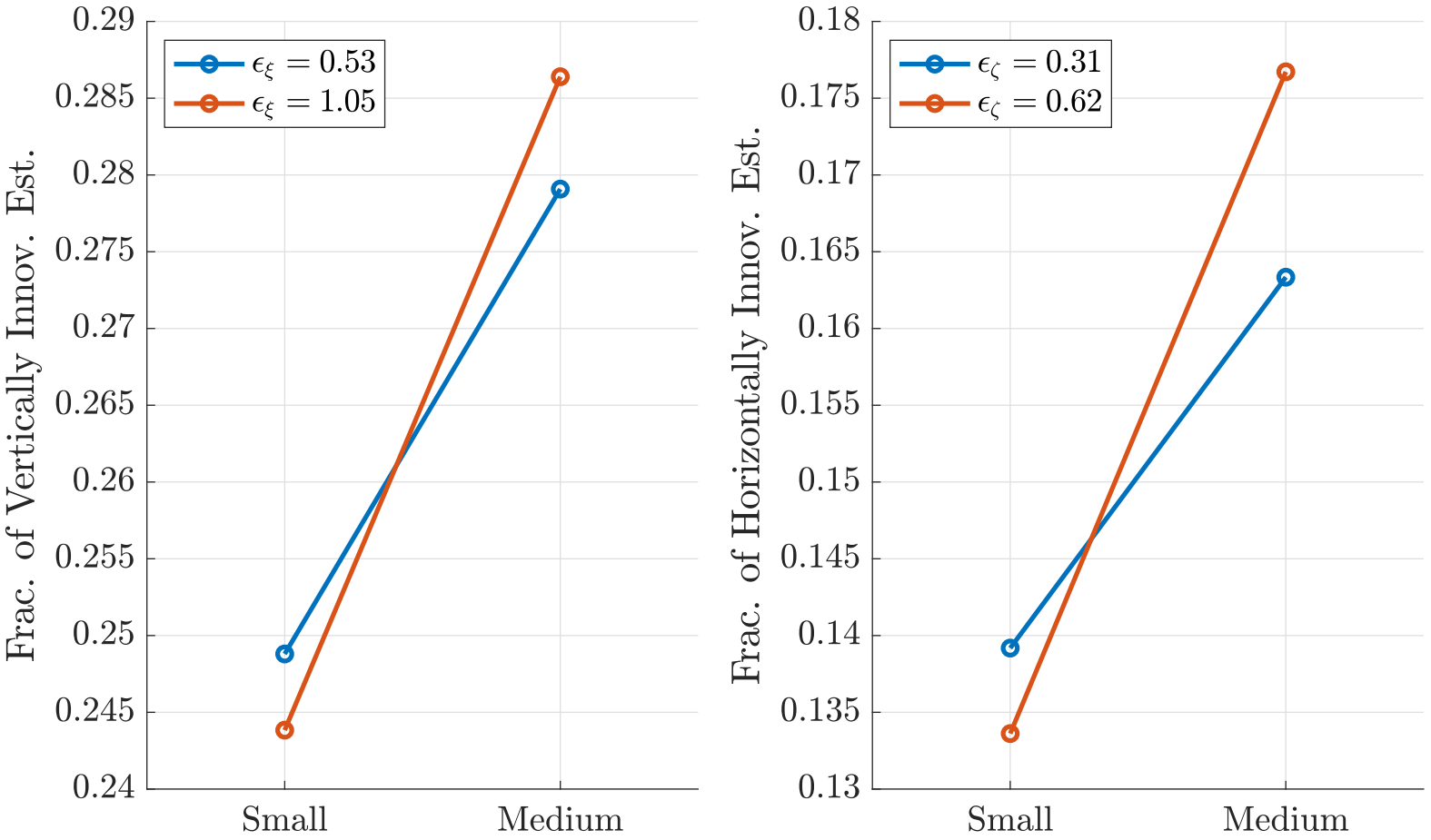
Model Fit

Dynamic Correlation with Innovation



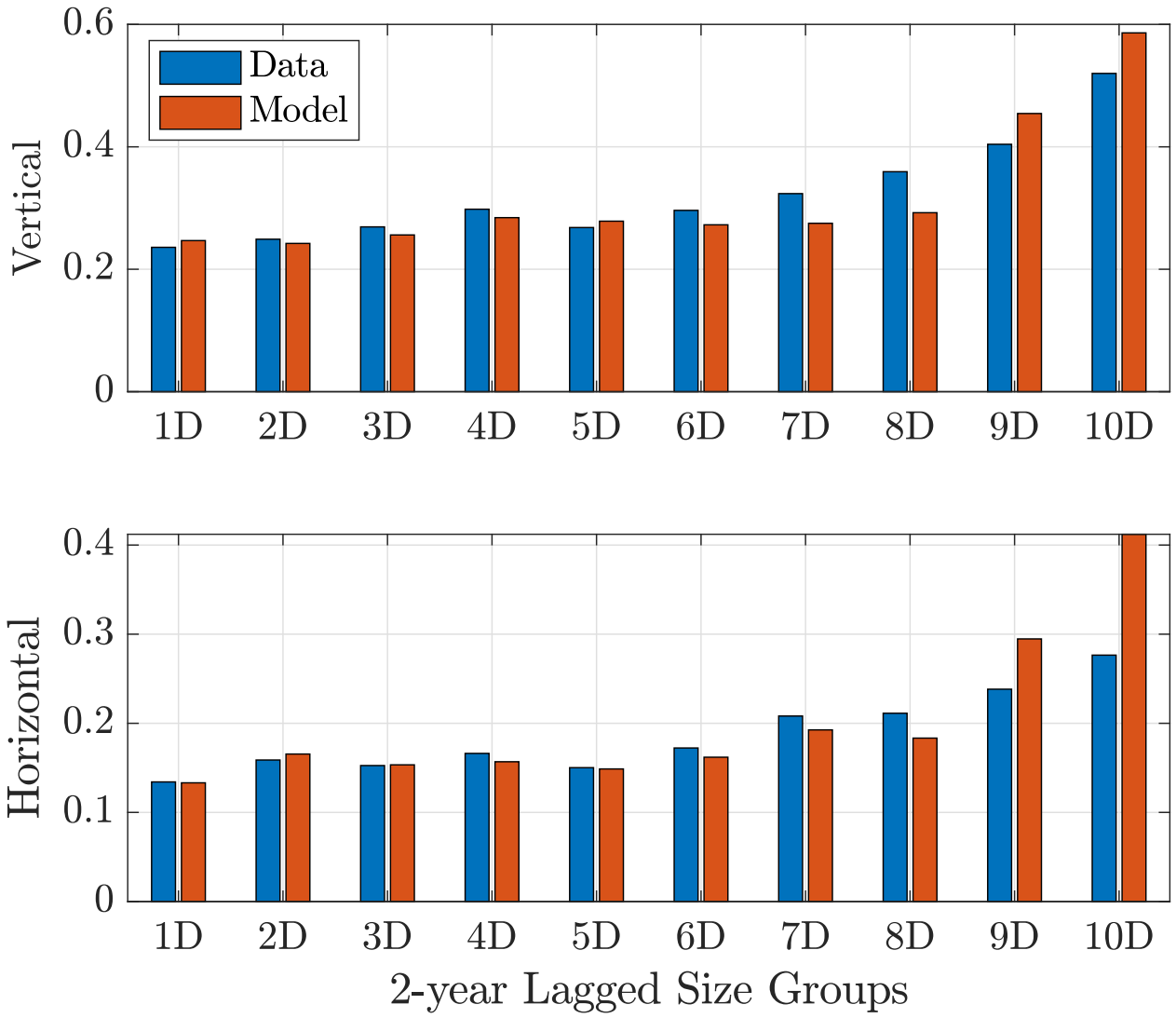
Identification of ϵ_ζ and ϵ_ξ

- A higher elasticity value leads to a steeper size profile of innovation frequency.



Model Fit on Larger Firms

Figure: Model Fit - Size dependence of Innovation Frequencies



Counterfactual Analysis

Counterfactual Analysis

- I focus on the Hartz Reform today.
  - Steady State with and without the policy introduction
    $\Rightarrow$ Impacts on growth and job turnover rates in S-S.
  - Transition from the baseline BGP to the new BGP after the policy introduction
    $\Rightarrow$ Welfare Analysis
- Three Major components of Hartz Reform (Jacobi and Kluve, 2006)
  1. improving employment services and policy measures
  2. activating the unemployed
  3. fostering employment demand by deregulating the labour market

Other Counterfactuals

The Hartz Reform: Aggregate Moments

- Reduction in labor wedge ω_L and firing cost parameter, κ.
- Changes in labor wedge to match the actual increase in employment.
- Decrease in κ to match the average firing cost.
- Frisch elasticity of labor supply = 1, entry elasticity = 0.1

Moment	Pre	$\omega_L = 0$	$\kappa = 0$	Post
Annual Output Growth Rate	1.23%	1.30%	1.23%	1.30%
Wage	1.0967	0.9993	1.0976	1.0000
R&D Labor Share	28.99%	27.23%	28.98%	27.22%
Avg. Annual Vertical Innov. Freq.	15.99%	16.85%	16.01%	16.86%
Avg. Annual Horizontal Innov. Freq.	0.52%	0.53%	0.52%	0.53%

Alternative Elasticity Values

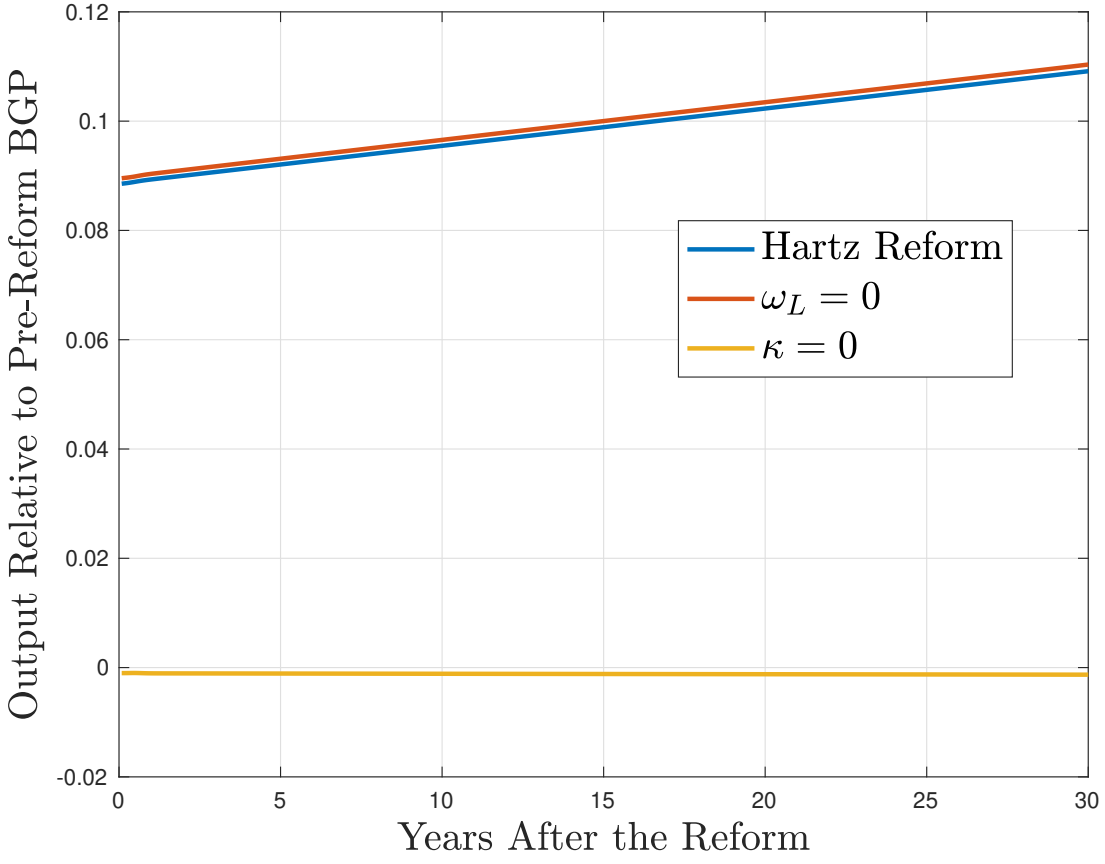
The Hartz Reform: Decomposition

- Almost all growth comes from vertical innovation.
- However, most job creation comes from horizontal and entrant innovations.

Moment	Pre	$\omega_L = 0$	$\kappa = 0$	Post
Verti. Innov. Contri. to Growth	91.04%	90.64%	91.28%	90.81%
Horiz. Innov. Contri. to Growth	2.68%	2.85%	2.61%	2.81%
Entr. Innov. Contri. to Growth	6.28%	6.52%	6.11%	6.39%
Job Creation Due to Verti. Innov.	22.76%	24.01%	23.12%	23.75%
Job Creation Due to Horiz. Innov.	27.45%	23.51%	26.30%	26.22%
Job Creation Due to Entr. Innov.	49.79%	52.48%	50.57%	50.03%
Job Dest. Due to Creative Dest.	32.71%	32.28%	28.20%	30.22%
Job Dest. Due to Obsolescence	62.89%	63.31%	67.22%	65.23%
Job Dest. Due to Exog. Exit	4.41%	4.41%	4.57%	4.55%

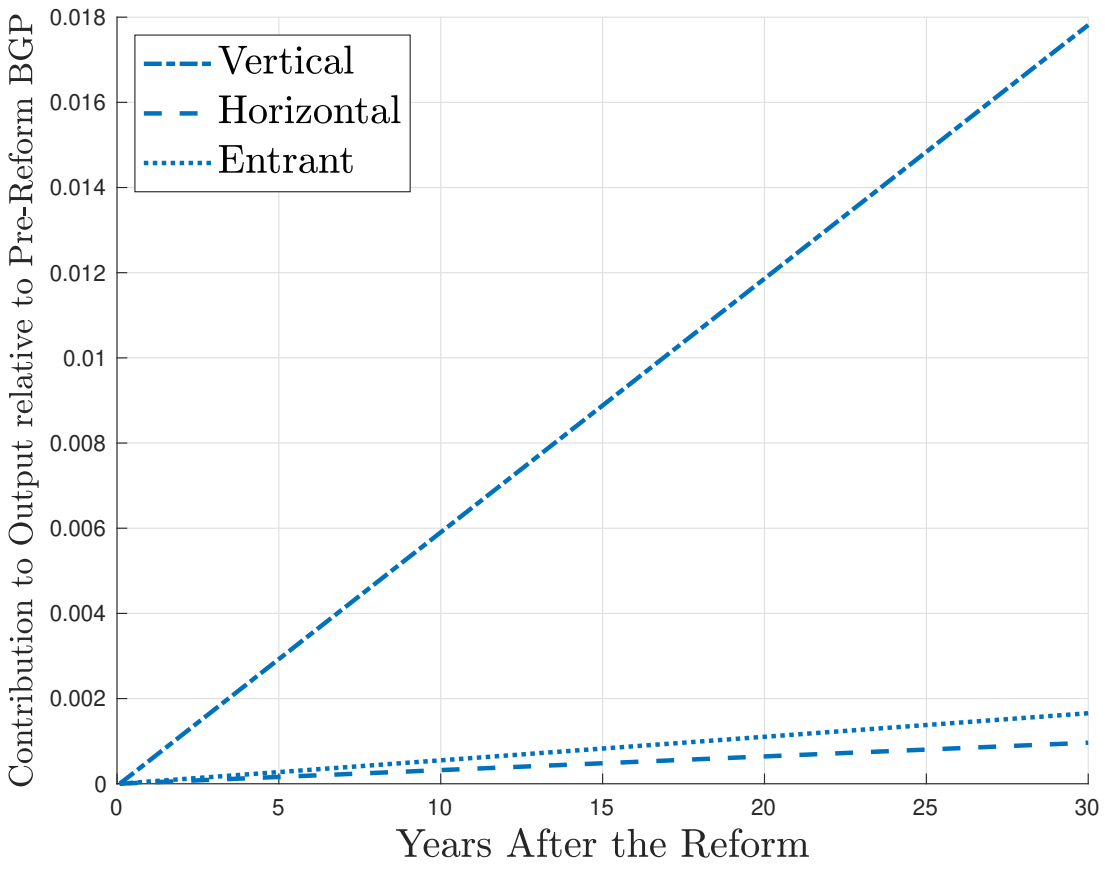
The Hartz Reform: Output in Transition and CEV

- The labor supply margin contributes to a static jump and faster growth.



The Hartz Reform: Output in Transition and CEV

- Vertical innovation makes a noticeable difference in productivity growth.



The Hartz Reform: Output in Transition and CEV

- Welfare gains from faster productivity growth are equivalent to 1% permanent increase in consumption.
- Vertical innovation accounts for 87% of all dynamic gains.

	$\omega_L = 0$	$\kappa = 0$	Post
CEV	6.23%	-0.14%	6.08%
$\Delta Y_0/Y_0^{BS}$	9.37%	-0.10%	9.26%
Δg_0^{verti}	0.86%	0.03%	0.88%
Δg_0^{horiz}	0.05%	-0.01%	0.05%
Δg_0^{entr}	0.11%	-0.03%	0.08%
ΔL	-3.37%	-0.04%	-3.41%

Hartz Exercise without Vertical Innovation

- The same post-reform calibration + new values for ω_L, κ
- Fix pre-reform vertical innovation rates as post-reform functions.
- Only horizontal and entrant innovations respond endogenously.

Moment	Baseline	Without V.I.
Δ Output Growth Rate	0.07ppt	0.01ppt
Δ Wage	-0.0973	-0.1059
Δ R&D Labor Share	-1.76ppt	-2.45ppt
Δ% Avg. Job Creation Rate	14.81%	12.26%
Δ% Avg. Job Destruction Rate	9.61%	8.13%
Δ% Avg. Vertical Innov. Freq.	65.42%	-0.43%
Δ% Avg. Horizontal Innov. Freq.	36.87%	39.15%

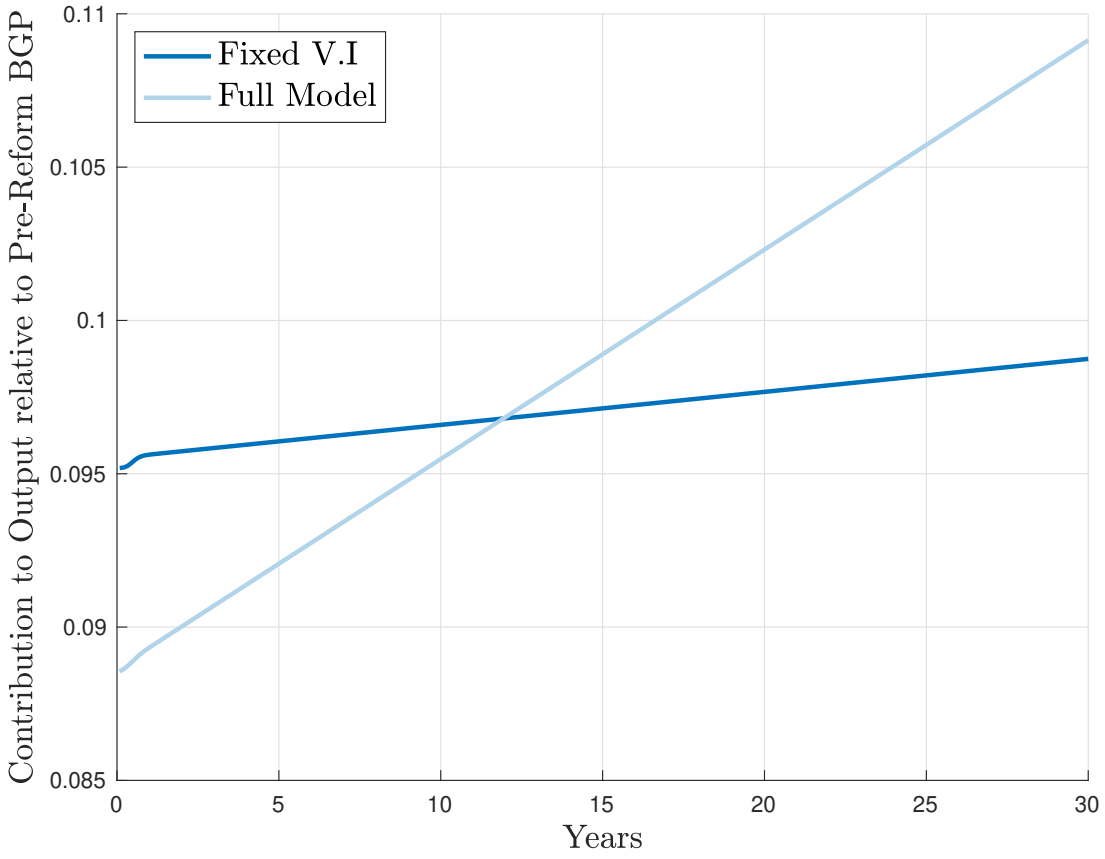
Decomposition without Vertical Innovation

- A large share of creative destruction in total job destruction process.

Moment	Baseline	Without V.I.
Δ Verti. Innov. Contri. to Growth	-0.21ppt	-0.81ppt
Δ Horiz. Innov. Contri. to Growth	0.12ppt	0.30ppt
Δ Entrant. Innov. Contri. to Growth	0.09ppt	0.52ppt
Δ Job Creation Due to Verti. Innov.	0.99ppt	-0.60ppt
Δ Job Creation Due to Horiz. Innov.	-1.23ppt	3.62ppt
Δ Job Creation Due to Entr. Innov.	0.24ppt	-3.02ppt
Δ Job Dest. Due to Creative Dest.	-2.49ppt	1.13ppt
Δ Job Dest. Due to Obsolescence	2.34ppt	-0.66ppt
Δ Job Dest. Due to Exog. Exit	0.14ppt	-0.47ppt

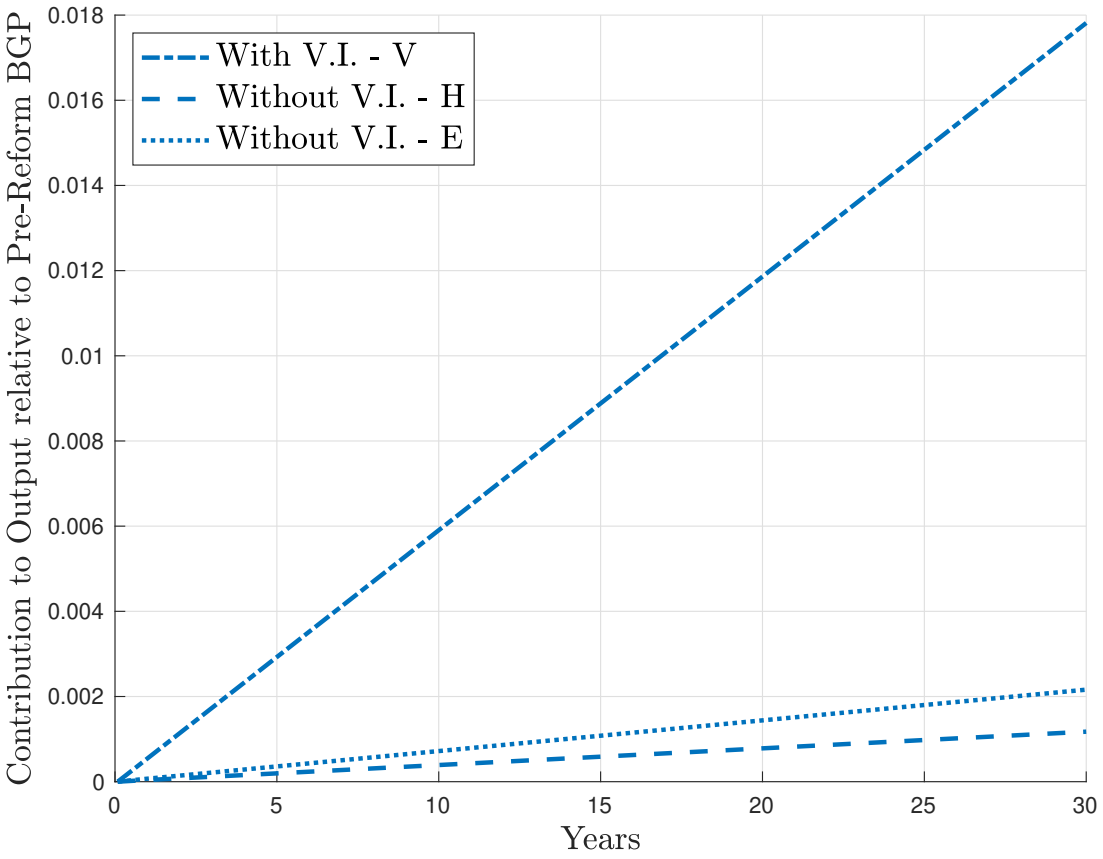
Transition Path without Vertical Innovation

- More initial jump, less productivity growth later.



Transition Path without Vertical Innovation

- Horizontal and entrant innovations not effective to boost productivity growth.



Transition Path without Vertical Innovation

- As a result, smaller welfare gains from the reform.
- Dynamic gains are underestimated by 84% without vertical innovation.

	Baseline	Without V.I.
CEV	6.08%	5.84%
$\Delta Y_0/Y_0^{BS}$	9.26%	9.99%
Δg_0^{verti}	0.88%	-0.01%
Δg_0^{horiz}	0.05%	0.06%
Δg_0^{entr}	0.08%	0.11%
ΔL	-3.41%	-3.43%

Concluding Remarks

- This paper proposes a new measure of establishment-level innovation activities for economic analysis.
- A quantitative model of vertical and horizontal innovation choice is constructed to encapsulate new facts established from the new measure.
  - Horizontal innovation generates more job flows.
  - However, vertical innovation accounts for most of economic growth.
- The model is used to quantify the welfare implications of labor market reform in Germany through incumbent innovation.
  - Labor market reform promotes not only a static output increase, but also a long-run productivity growth.
  - Most welfare and growth gains come from additional vertical innovation.
  - Incumbents' vertical innovation accounts for 86% of welfare gains from future output growth.

Appendix

Variables in IAB-BP

- The survey has information on extensive establishment-level variables
  - Employment as of June 30, # of hiring in the fist half of the year, # of separation in the first half of the year, Labor expenses as of June, Investment last year, Business volume last year, # of training participation in the first half of the year
- Establishments are asked whether they implemented the following innovation activities in the previous year.
  - Process innovation: improve the production process
  - Quality improvement: improve the products they previously produced
  - Product expansion: start to offer a product that previously existed in the market
  - Variety creation: start to offer a product that did not exist in the market
- Vertical innovation includes process innovation and quality improvement while horizontal innovation includes product expansion.

Definitions of Outcome Variables

- Definitions of Outcome Variables
  - Employment growth = $(AdmEmp - L.AdmEmp)/(\frac{1}{2}AdmEmp + \frac{1}{2}L.AdmEmp)$
  - Wage, Revenue, Investment growth rates are defined similarly.
  - Hiring rate = $2 \times Hiring/(\frac{1}{2}AdmEmp + \frac{1}{2}L.AdmEmp)$
  - Separation rate = $2 \times Separation/(\frac{1}{2}AdmEmp + \frac{1}{2}L.AdmEmp)$
  - Training Rate = $2 \times \#ClassParticipation/(\frac{1}{2}AdmEmp + \frac{1}{2}L.AdmEmp)$
  - Establishment age = Survey Year - Year of Foundation

Definitions of Outcome Variables - Continued

- The variables are further defined by the following.
  - $AdmEmp$ = Employment of an establishment as of June 30 in the survey year from the administrative data
  - $Hiring$ = Total number of newly hired workers in the first half of the survey year from the survey
  - $Separataion$ = Total number of separated workers in the first half of the survey year from the survey
  - $Wage$ = Imputed mean daily wage as of June in the survey year from the administrative data
  - $Revenue$ = Business volume in the previous year from the survey
  - $Investment$ = Sum of investment expenditures in the previous year from the survey
  - $\#ClassParticipation$ = Total number of participants or cases of participation in internal or external training courses the establishments cover the expenses in full or in part

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Summary Statistics for Outcome Variables

- All results are weighted by cross-sectional weights.

Table: Summary Statistics - Outcome Variables (in %)

	Mean	Sd	Median	1st Q.	3rd Q.	N.
Employment Gr.	0.72	22.06	0.00	0.00	4.88	110,946
Hiring Rate	13.77	40.56	0.00	0.00	10.53	99,046
Separation Rate	11.39	36.18	0.00	0.00	6.30	99,112
Wage Gr.	2.14	14.65	1.27	-1.52	5.93	91,358
Rev. Gr.	2.13	27.86	0.00	-7.68	12.39	77,919
Invest. Gr.	-5.63	1.55	0.00	-198.01	169.23	81,008
Training Rate	79.73	184.23	24.29	0.00	131.58	85,546

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Dynamic Correlation - Specification

- Variables of interest, Y_{it}:
  - Employment growth rate $= (Emp - L.Emp)/(\frac{1}{2}Emp + \frac{1}{2}L.Emp)$
  - Wage growth rate $= (Wage - L.Wage)/(\frac{1}{2}Wage + \frac{1}{2}L.Wage)$
  - Employment and wage variables come from the administrative data.

Other Variables Descriptive Statistics

- Consider the following dynamic correlation model.

$$Y_{it} = \sum_{s=-5}^{5} \mathbf{1}_{it+s}^{a} \beta_s^a + \gamma' X_{it} + \varepsilon_{it}$$

where $\mathbf{1}_{it}^{a}$ is indicator of innovation type $a =$ vertical or horizontal.

- Controls: Other types of innovation, establishment and year f.e..

! Entrant/exiting firms are excluded.

Summary of Establishment-level Findings

- Vertical innovation
  - 1.05% employment increase in the innovation year.
  - The increment in employment depreciates fast and becomes insignificant three years after (-0.93%).
  - Short-run wage increase by 0.65%.
- Horizontal innovation
  - Similar 1.23% increase in employment over the innovation year.
  - This effect accumulates and becomes larger three years after (3.25%).
  - No wage increase.

- Vertical innov. $\Rightarrow$ within-line productivity $\Uparrow$ $\Rightarrow$ small size $\Uparrow$, depreciates fast
- Horizontal innov. $\Rightarrow$ additional product line $\Rightarrow$ large size $\Uparrow$, long-lived

Results in Table Results in Graph

Innovation and Non-Innovating Establishments

- The panel regression specification, $j(i)$ for sector and t for year

$$Y_{it} = \beta^V L.V_{j(i)t} + \beta^H L.H_{j(i)t} + \gamma' X_{it} + \epsilon_{it},$$

Controls include fixed effects, lagged size, HHI, industrial emp. growth.

	Emp. Gr.	Exit	Wage Gr.	Emp. Gr.	Exit	Wage Gr.
$V_{j(i)t}$	1.45	0.49	-0.37	3.34	1.21	1.04
	(2.88)	(1.78)	(1.71)	(2.92)	(1.82)	(1.74)
$H_{j(i)t}$	-10.06***	6.10***	5.89***	-10.85***	5.49***	5.24***
	(2.88)	(1.78)	(1.71)	(2.92)	(1.82)	(1.74)
$E_{j(i)t}$				-38.85**	33.97***	-23.52**
				(19.11)	(11.91)	(11.82)
N.	20,997	21,921	18,024	20,997	21,149	18,024
F.E.	Year	Year	Year	Year	Year	Year
Ind. Ctr.	No	No	No	Yes	Yes	Yes

- Horizontal innov. has stronger impact on job turnover than vertical innov.

Results in Table Comparison to Patent Data

Regression Results - Employment Growth

- 1 Year - β_0^a, 4 Year - $\sum_{s=0}^{3} \beta_s^a$, $a =$ vertical or horizontal.

N. Lags	1 Year	1 Year	1 Year	4 Years	4 Years	4 Years
Vertical Innov.	1.05^{***}		0.85^{***}	-0.34		-0.93
	(0.31)		(0.32)	(0.67)		(0.75)
Horizontal Innov.		1.23^{***}	1.10^{***}		2.42^{***}	3.25^{***}
		(0.35)	(0.38)		(0.83)	(0.97)
N.	100,770	100,770	100,770	100,770	100,770	100,770
Other Innov.			✓			✓
Fixed Effects	Est.	Est.	Est.	Est.	Est.	Est.

Regression Results - Hiring Rate

- 1 Year - β_0^a, 4 Year - $\sum_{s=0}^{3} \beta_s^a$, $a =$ vertical or horizontal.

N. Lags	1 Year	1 Year	1 Year	4 Years	4 Years	4 Years
Vertical Innov.	1.98^{***}		1.26^{**}	2.33^{*}		0.97
	(0.59)		(0.62)	(1.28)		(1.44)
Horizontal Innov.		2.75^{***}	2.41^{***}		3.80^{**}	3.34^{*}
		(0.68)	(0.73)		(1.60)	(1.86)
N.	100,406	100,406	100,406	100,406	100,406	100,406
Other Innov.			✓			✓
Fixed Effects	Est.	Est.	Est.	Est.	Est.	Est.

Regression Results - Separation Rate

- 1 Year - β_0^a, 4 Year - $\sum_{s=0}^{3} \beta_s^a$, $a =$ vertical or horizontal.

N. Lags	1 Year	1 Year	1 Year	4 Years	4 Years	4 Years
Vertical Innov.	0.56		0.28	2.96**		2.61*
	(0.59)		(0.62)	(1.28)		(1.45)
Horizontal Innov.		0.76	0.62		1.28	-0.09
		(0.68)	(0.73)		(1.60)	(1.86)
N.	100,475	100,475	100,475	100,475	100,475	100,475
Other Innov.			✓			✓
Fixed Effects	Est.	Est.	Est.	Est.	Est.	Est.

Regression Results - Wage Growth

- 1 Year - β_0^a, 4 Year - $\sum_{s=0}^{3} \beta_s^a$, $a =$ vertical or horizontal.

N. Lags	1 Year	1 Year	1 Year	4 Years	4 Years	4 Years
Vertical Innov.	0.60***		0.65***	-0.79**		-0.89**
	(0.17)		(0.18)	(0.37)		(0.41)
Horizontal Innov.		-0.10	-0.31		-0.18	0.05
		(0.20)	(0.21)		(0.46)	(0.53)
N.	91,482	91,482	91,482	91,482	91,482	91,482
Other Innov.			✓			✓
Fixed Effects	Est.	Est.	Est.	Est.	Est.	Est.

Regression Results - Revenue Growth

- 1 Year - β_0^a, 4 Year - $\sum_{s=0}^{3} \beta_s^a$, $a =$ vertical or horizontal.

N. Lags	1 Year	1 Year	1 Year	4 Years	4 Years	4 Years
Vertical Innov.	1.33***		0.98**	-2.34***		-2.70***
	(0.36)		(0.38)	(0.78)		(0.87)
Horizontal Innov.		1.27***	0.86**		0.36	1.12
		(0.41)	(0.44)		(0.97)	(1.12)
N.	71,318	71,318	71,318	71,318	71,318	71,318
Other Innov.			✓			✓
Fixed Effects	Est.	Est.	Est.	Est.	Est.	Est.

Go back

Regression Results - Investment Growth

- 1 Year - β_0^a, 4 Year - $\sum_{s=0}^{3} \beta_s^a$, $a =$ vertical or horizontal.

N. Lags	1 Year	1 Year	1 Year	4 Years	4 Years	4 Years
Vertical Innov.	29.06^{***}		24.88^{***}	12.37^{***}		8.49^{*}
	(2.09)		(2.20)	(4.53)		(5.09)
Horizontal Innov.		25.00^{***}	15.64^{***}		18.51^{***}	12.77^{*}
		(2.38)	(2.55)		(5.66)	(6.54)
N.	73,851	73,851	73,851	73,851	73,851	73,851
Other Innov.			✓			✓
Fixed Effects	Est.	Est.	Est.	Est.	Est.	Est.

Regression Results - Training Rate

- 1 Year - β_0^a, 4 Year - $\sum_{s=0}^{3} \beta_s^a$, $a =$ vertical or horizontal.

N. Lags	1 Year	1 Year	1 Year	4 Years	4 Years	4 Years
Vertical Innov.	26.89***		15.92*	36.22**		-0.37
	(7.88)		(8.36)	(18.21)		(20.66)
Horizontal Innov.		37.80***	27.49***		74.46***	48.87*
		(9.03)	(9.78)		(22.71)	(26.56)
N.	86,869	86,869	86,869	86,869	86,869	86,869
Other Innov.			✓			✓
Fixed Effects	Est.	Est.	Est.	Est.	Est.	Est.

Go back

Industry-level Innovation and Non-Innovating Establishments

- The panel regression specification, $j(i)$ for sector and t for year

$$Y_{it} = \beta^V L.V_{j(i)t} + \beta^H L.H_{j(i)t} + \gamma' X_{it} + \epsilon_{it},$$

Controls include fixed effects, lagged size, HHI, industrial emp. growth.

	Dep. Variable: Employment Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
Vertical Innov.	-2.03		3.34	-24.17***		-17.35
	(2.57)		(2.92)	(8.89)		(9.28)
Horizontal Innov.		-9.34***	-10.85***		-32.05***	-29.51***
		(2.57)	(2.92)		(9.60)	(10.13)
Entrant Emp. Share			-38.85**			-53.71
			(19.11)			(30.03)
N.	20,997	20,997	20,997	20,839	20,839	20,839
Fixed Effects	Year	Year	Year	Year & Sector	Year & Sector	Year & Sector

Go back

Industry-level Innovation and Non-Innovating Establishments

- The panel regression specification, $j(i)$ for sector and t for year

$$Y_{it} = \beta^V L.V_{j(i)t} + \beta^H L.H_{j(i)t} + \gamma' X_{it} + \epsilon_{it},$$

Controls include fixed effects, lagged size, HHI, industrial emp. growth.

	Dep. Variable: Establishment Exit					
	(1)	(2)	(3)	(4)	(5)	(6)
Vertical Innov.	4.00**		1.21	6.11		9.66
	(1.60)		(1.82)	(5.44)		(5.68)
Horizontal Innov.		6.15***	5.49***		-10.34	-12.79**
		(1.60)	(1.82)		(5.89)	(6.21)
Entrant Emp. Share			33.97***			10.06
			(11.91)			(18.41)
N.	21,149	21,149	21,149	20,989	20,989	20,989
Fixed Effects	Year	Year	Year	Year & Sector	Year & Sector	Year & Sector

Go back

Industry-level Innovation and Non-Innovating Establishments

- The panel regression specification, $j(i)$ for sector and t for year

$$Y_{it} = \beta^V L.V_{j(i)t} + \beta^H L.H_{j(i)t} + \gamma' X_{it} + \epsilon_{it},$$

Controls include fixed effects, lagged size, HHI, industrial emp. growth.

	Dep. Variable: Wage Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
Vertical Innov.	3.64**		1.04	4.87		1.71
	(1.44)		(1.69)	(5.18)		(5.40)
Horizontal Innov.		5.81***	5.24***		11.97**	9.75
		(1.48)	(1.74)		(5.62)	(5.93)
Entrant Emp. Share			-23.52**			-32.68
			(11.82)			(17.87)
N.	18,024	18,024	18,024	17,878	17,878	17,878
Fixed Effects	Year	Year	Year	Year & Sector	Year & Sector	Year & Sector

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Comparison between Survey and Patent Measures of Innovation

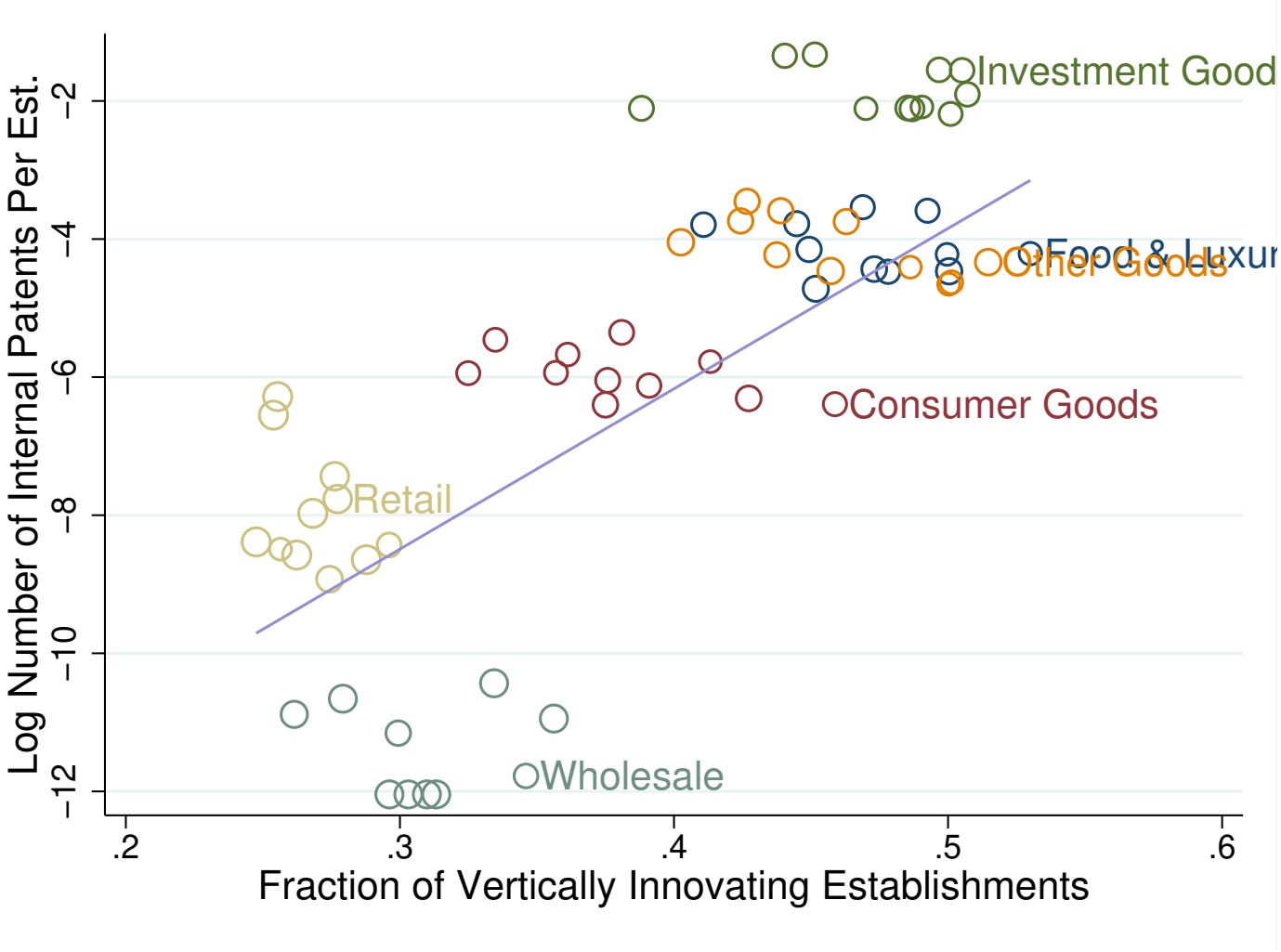
- Construct industry-year level innovation measures using the number of patents granted to German entities.
- Distinction between internal vs. external innovation (Akcigit and Kerr, 2018)
  - **Internal Innovation**: self-citation share ≥ 0.5
  - **External Innovation**: self-citation share < 0.5
- The number of patents does not even exist for agriculture, mining, construction, and service industries.
- Overall survey-based fraction of innovating establishments are strongly positively correlated with the number of patents across different industries and years.

Overall Innovation

- Fraction of either vertically or horizontally innovating establishments vs. the total number of patents

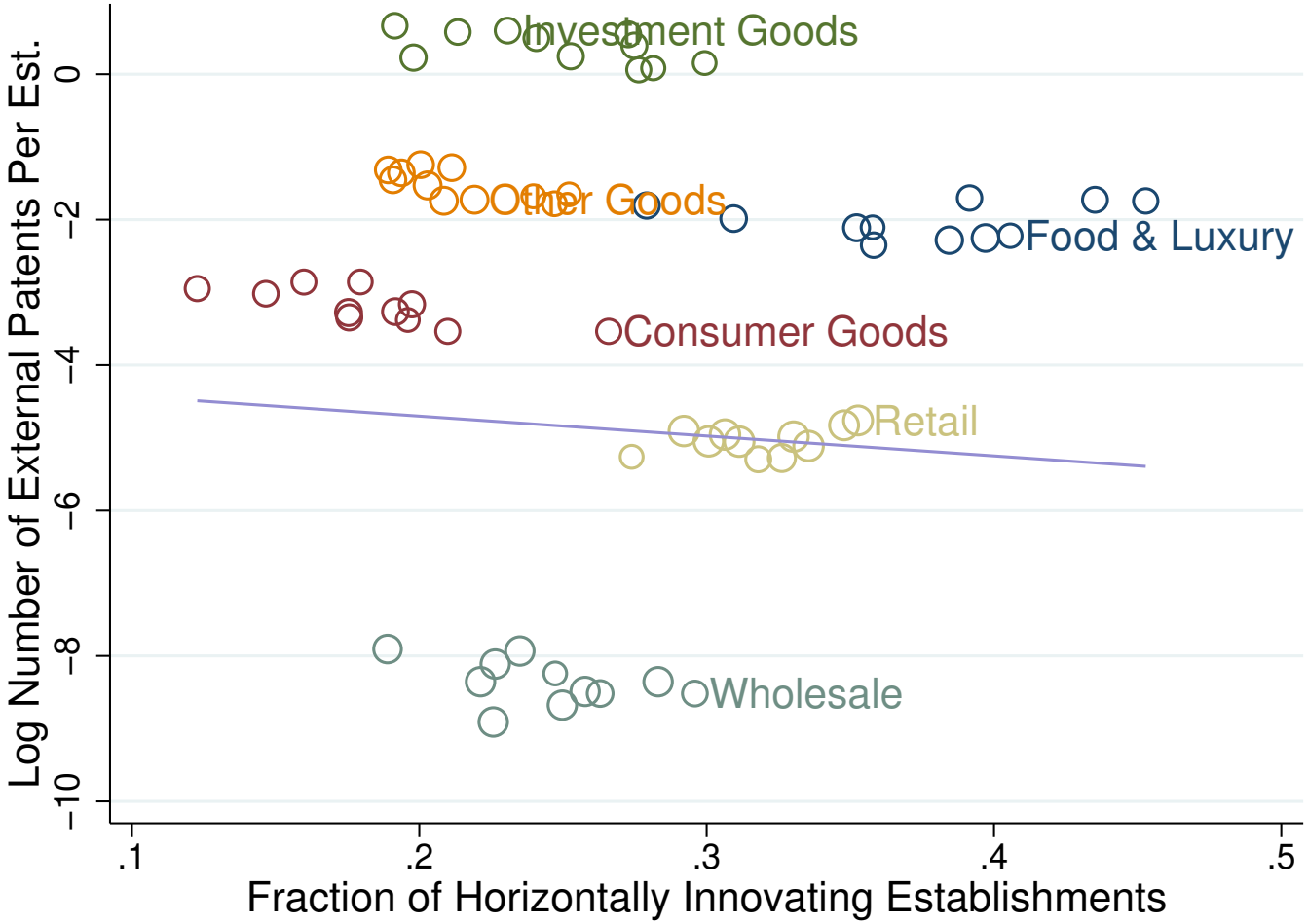
Vertical Innovation vs. Internal Innovation

- Fraction of vertically innovating establishments vs. the total number of internal patents

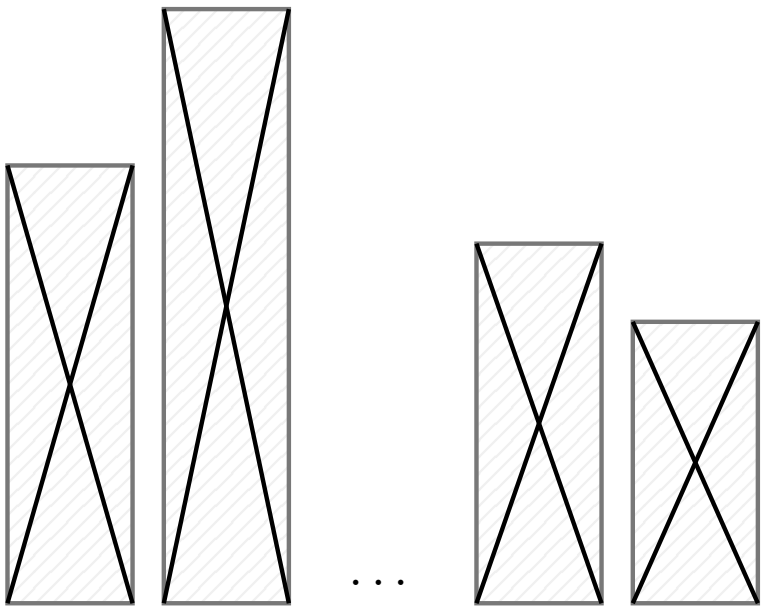


Horizontal Innovation vs. External Innovation

- Fraction of horizontally innovating establishments vs. the total number of external patents

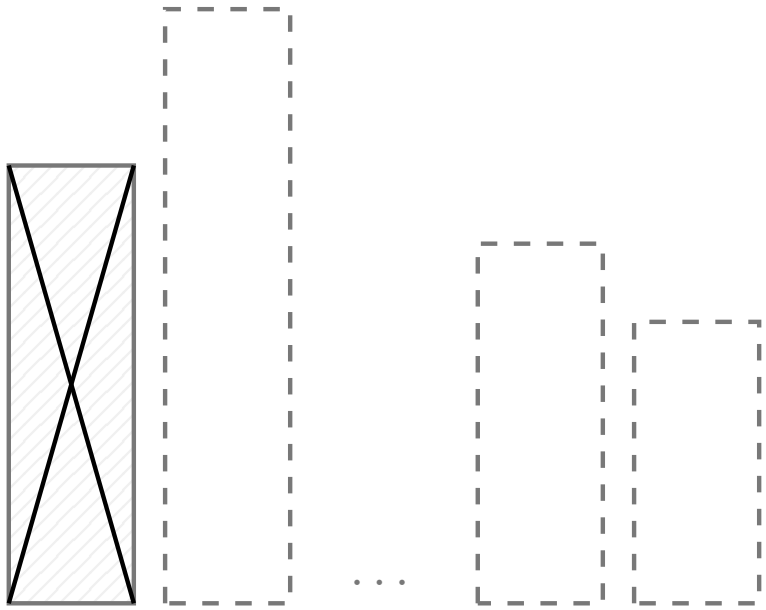


Graphics: Firm Exit



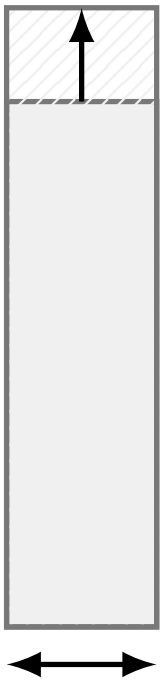
- A firm exits the market in two cases.
  1. Exogenously with rate δ.
  $\Rightarrow$ Product lines held by exiting firms are reallocated toward expanding & entering firms.
  2. The firm loses the last product line by creative destruction.

Graphics: Firm Exit



- A firm exits the market in two cases.
  1. Exogenously with rate δ.
  $\Rightarrow$ Product lines held by exiting firms are reallocated toward expanding & entering firms.
  2. The firm loses the last product line by creative destruction.

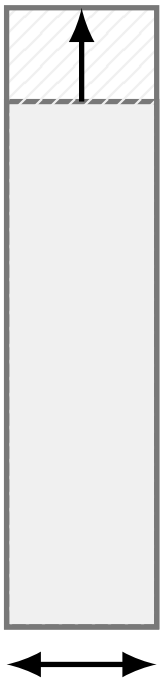
Graphics: Firm Entry



$$\Delta x = \lambda^e \bar{x}$$

- An entering firm is undirectedly matched to a product line to learn and makes vertical innovation with a constant step size $\lambda^e \bar{x}$.
- With probability p, it is matched to a vacant product line.

Graphics: Firm Entry

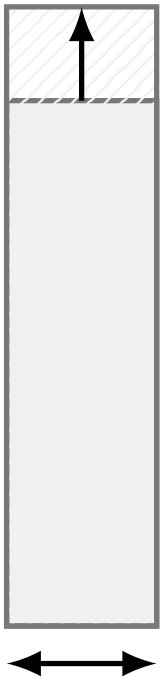


$$\Delta x = \lambda^e \bar{x}$$

- An entering firm is undirectedly matched to a product line to learn and makes vertical innovation with a constant step size $\lambda^e \bar{x}$.
- With probability p, it is matched to a vacant product line.

Go back

Graphics: Firm Entry



$$\Delta x = \lambda^e \bar{x}$$

- An entering firm is undirectedly matched to a product line to learn and makes vertical innovation with a constant step size $\lambda^e \bar{x}$.
- With probability $1 - p$, it is matched to an occupied product line.
- It replaces the existing firm only if it is more productive.

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Distribution

- The distribution of product lines across x satisfies

$$g_x x f(x) = \delta F(x) + \int_{s \in (x - \lambda^i \bar{x}, x]} \xi(s) dF(s) + (1-p)(G(\infty) - G(x))F(x) - pG(x)$$

  where M is the number of the product lines.

- The distribution of expanding product lines is characterized by

$$G(x) = \int_{s \leq x} \zeta(s) dF(s) + \int_{s \leq x - \lambda^e \bar{x}} \chi dF(s).$$

- $\bar{x} := \int x dF(x) = \int_\Omega x_i di = 1.$

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Creative Destruction

- The creative destruction rate is then

$$\tau(x) = (1-p)\left(\chi\left(\int_{s>x-\lambda^e\bar{x}} dF(s)\right) + \int_{s>x} \zeta(s)dF(s)\right).$$

- The probability of being matched to a vacant line, p, is characterized by

$$\delta = Mp\int \zeta(s)dF(s) + \chi p.$$

Go back

Free Entry and Labor Market Clearing

- The free entry condition is given by

$$wc_{\chi}(\chi) = \chi \left(\int (p + (1-p)F(x + \lambda^e \bar{x}))v(x + \lambda^e \bar{x}) dF(x) \right)$$

  where $c_{\chi}(\chi)/\chi$ is increasing in χ.

- The labor market clearing condition is expressed as

$$wL = \frac{\sigma - 1}{\sigma} + w \int_0^1 c_{\xi}(\xi_{jt}) dj + w \int_0^1 c_{\zeta}(\zeta_{jt}) dj + wc_{\chi}(\chi)$$

Go back

Model Fit for Untargeted Moments

Moment	Data	Model
Corr. Coeff. between Vertical and Horizontal Innov.	0.38	0.23
2-Year Growth Rate Diff. across Size 3T and 1T	-0.15ppt	0.71ppt
2-Year Growth Rate Diff. across Size 2T and 1T	-0.18ppt	0.36ppt
Job Destruction Rate Diff. across Size 3T and 1T (Surveyed)	-2.72ppt	-0.09ppt
Job Destruction Rate Diff. across Size 2T and 1T (Surveyed)	-1.38ppt	0.02ppt
Job Creation Rate (Surveyed)	8.22%	0.68%
Job Destruction Rate (Surveyed)	7.51%	0.60%

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Alternative Elasticity Values

	Baseline	$\eta = 0.3$	$\eta = 4$	$\epsilon_\chi = 0$	$\epsilon_\chi = 1$
ΔGrowth	0.07%p	0.07%p	0.07%p	0.07%p	0.12%p
ΔWage	-0.0974	-0.0962	-0.0969	-0.0979	-0.0911
ΔR&D Share	-1.81%p	-1.78%p	-1.80%p	0.34%p	-2.01%p
ΔJC	0.12%p	0.12%p	0.12%p	0.14%p	0.15%p
ΔJD	0.10%p	0.10%p	0.10%p	0.13%p	0.10%p
ΔVertical	0.82%p	0.81%p	0.82%p	0.82%p	0.80%p
ΔHorizontal	0.01%p	0.01%p	0.01%p	0.02%p	0.01%p
CEV	6.152%	6.305%	6.012%	6.806%	6.533%
dY_0/Y_0^{BS}	9.272%	9.158%	9.230%	9.330%	9.159%
Δg_0^{verti}	0.858%	0.848%	0.854%	0.859%	1.248%
Δg_0^{horiz}	0.066%	0.065%	0.065%	0.046%	0.190%
Δg_0^{entr}	0.129%	0.127%	0.127%	0.072%	0.502%
ΔL	-3.400%	-3.140%	-3.493%	-2.698%	-3.530%

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R&D Tax Incentive: Aggregate Moments

- 25% tax exemption of R&D expenditures ≈ 4.41% of subsidy rate

Moment	Baseline	$s_\xi > 0$	$s_\zeta > 0$	$s_\chi > 0$	All
Output Growth Rate	1.30%	1.33%	1.30%	1.30%	1.33%
Wage	1.0000	1.0018	1.0013	1.0054	1.0076
R&D Labor Share	27.24%	27.47%	27.39%	27.05%	27.45%
Avg. Monthly Vertical Innov. Freq.	1.41%	1.44%	1.41%	1.40%	1.44%
Avg. Monthly Horizontal Innov. Freq.	1.42%	1.42%	1.44%	1.42%	1.44%
Verti. Innov. Contri. to Growth	90.81%	90.99%	90.57%	90.70%	90.66%
Horiz. Innov. Contri. to Growth	2.81%	2.75%	2.91%	2.83%	2.87%
Entr. Innov. Contri. to Growth	6.38%	6.26%	6.52%	6.47%	6.47%
Job Creation Due to Verti. Innov.	24.03%	25.26%	24.53%	23.44%	23.93%
Job Creation Due to Horiz. Innov.	23.62%	21.95%	23.06%	26.64%	26.90%
Job Creation Due to Entr. Innov.	52.36%	52.79%	52.40%	49.93%	49.17%
Job Dest. Due to Creative Dest.	28.18%	33.60%	30.02%	31.40%	27.38%
Job Dest. Due to Obsolescence	66.35%	61.44%	65.31%	64.15%	67.64%
Job Dest. Due to Exog. Exit	5.48%	4.96%	4.67%	4.45%	4.98%

R&D Tax Incentive: Aggregate Moments

- 25% tax exemption of R&D expenditures ≈ 4.41% of subsidy rate

Moment	$s_\xi > 0$	$s_\zeta > 0$	$s_\chi > 0$	All
CEV	0.20%	-0.02%	-0.70%	-0.53%
$\Delta Y_0/Y_0^{BS}$	-0.14%	-0.07%	-0.49%	-0.69%
Δg_0^{verti}	0.39%	0.01%	-0.04%	0.36%
Δg_0^{horiz}	-0.00%	0.02%	0.00%	0.02%
Δg_0^{entr}	0.00%	0.03%	0.02%	0.03%
ΔL	-0.05%	-0.02%	-0.19%	-0.27%
$t \vert Y_t^{TR} = Y_t^{BS}$ in years	5.34	14.40	-	24.74

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